2CB03 26

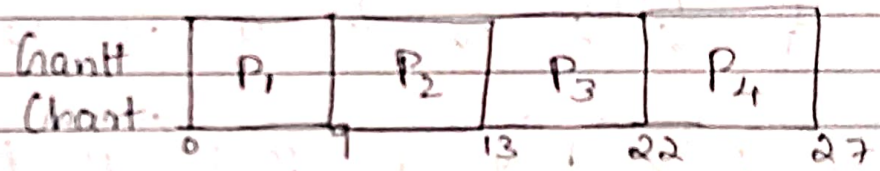
D D M M Y Y Y Y

Module - 2.

FCFS

4)a)

| Process | AT | BT | CT | TAT | WT |
|---|---|---|---|---|---|
| $P_1$ | 0 | 9 | 9 | 9 | 0 |
| $P_2$ | 1 | 4 | 13 | 12 | 8 |
| $P_3$ | 2 | 9 | 22 | 20 | 11 |
| $P_4$ | 3 | 5 | 27 | 24 | 19 |
| | | | | 16.25 | 19 |




TAT = CT − AT

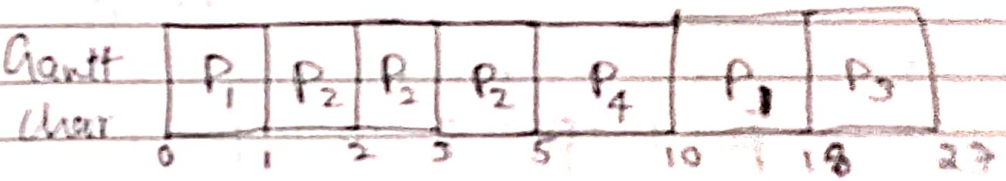
WT = TAT − BT

Avg TAT = 16.25 ms

Avg WT = 19 ms

SRJF

| Process | AT | BT | CT | TAT | WT |
|---|---|---|---|---|---|
| $P_1$ | 0 | 9 8 7 | 18 | 18 | 9 |
| $P_2$ | 1 | 4 3 2 | 5 | 4 | 0 |
| $P_3$ | 2 | 9 | 27 | 25 | 16 |
| $P_4$ | 3 | 5 | 10 | 7 | 2 |
| | | | | 13.25 | 5.75 |




Avg TAT = 13.5

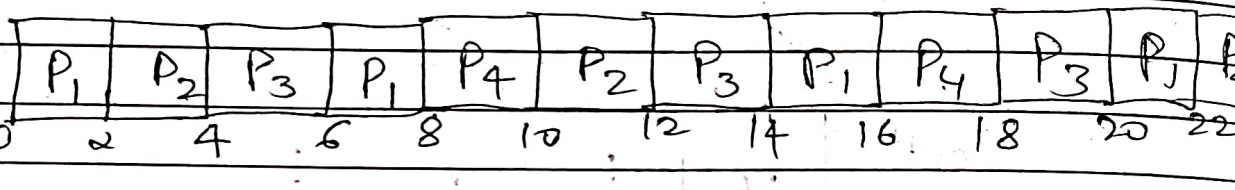
Avg WT = 6.75

DD MM YYYY

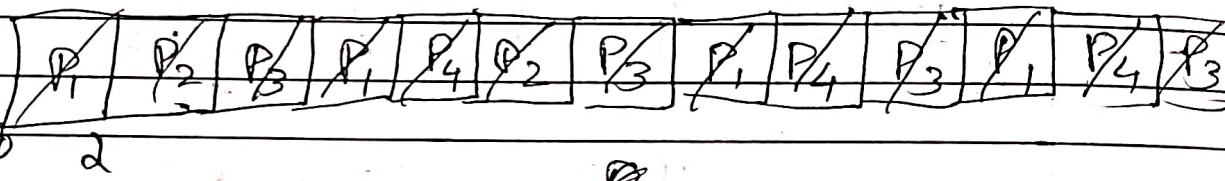
RR ( q = 2ns)

| | AT | BT | CT | TAT | WT |
|---|---|---|---|---|---|
| $P_1$ | 0 | 8 | 26 | 26 | 17 |
| $P_2$ | 1 | 4 | 12 | 11 | 7 |
| $P_3$ | 2 | 9 | 27 | 25 | 16 |
| $P_4$ | 3 | 5 | 23 | 20 | 15 |

Gantt Chart




Ready q



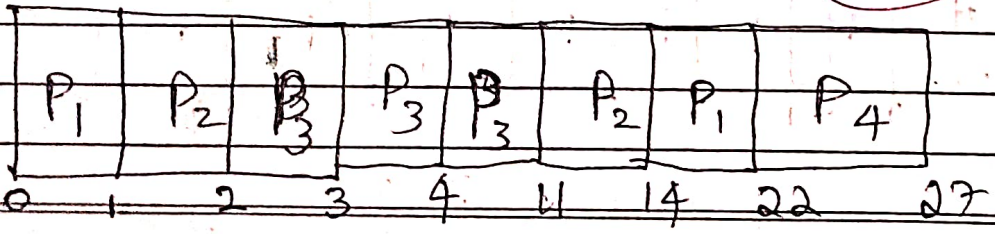

Avg TAT = 20.5

Avg WT = 13.75

Priority lower no ↑ P

| Prori | | AT | BT | CT | TAT | WT |
|---|---|---|---|---|---|---|
| 3 | $P_1$ | 0 | 8 | 22 | 22 | 13 |
| 2 | $P_2$ | 1 | 3 | 14 | 13 | 9 |
| ⇒ 1 | $P_3$ | 2 | 7 | 11 | 9 | 0 |
| 4 | $P_4$ | 4 | 5 | 27 | 23 | 19 |
| | | | | | 16.75 | 10.25 |

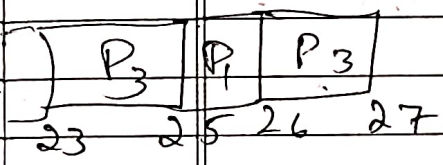
Gantt Chart

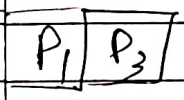


D D M M Y Y Y Y

Avg TAT = 16.25

Avg WT = 10.25




1) b)

| Process | Thread |
|---|---|
| → Process has separate memory block assigne to it | → All thread share common memory blo |
| → Procss take long time to excute | → Threads take relativel less time to excu |
| → Procss in less efficent | → Thread in more effic |
| → Procss contain listy Sub task to excute | → Thread contains 1 task to excute at once. |
| → Process can be braked into multiple thread | → Thread is small unit of Process. |

D D M M Y Y Y Y

Module - 1

2) b) System calls

System calls are the interface through which process get the resource resources from O/System (OS).

Types
- → Process control
- → File Management
- → Device Mangm
- → Information mangm
- → Communication mgm
- → Protection

(i) Process control

System call include
- end, abort
- load, execute
- create process, terminate process
- wait for time
- wait event, signal event
- allocate and deallocate memory

→ using above system calls process can be created, launched, monitored, paused, resumed, terminated

D D M M Y Y Y Y

→ When child process is executing, parent process wait
→ When 1 process is at wait, another execute.
→ Allocation & deallocation of memory when created & deleted.

(2) File Management:

System call include: create a file, delete a file, open a file, close a file, read, write, reposition, get/set file attribute.

→ A file can be ~~open~~ created, opened and read / write, the reposition of file pointer is done.

→ file attributes like file name, file size, file type can be accessed

(3) Device Mangement

System call include: request ~~allocate~~ a device, assign ~~deallocate~~ device, read, write, reposition, get/set device attribute. logically allocate & deallocate.

When a process require device to further proceed, it makes these call

D D M M Y Y Y Y

→ When process request a device, if the device is in available state it is assigned to process else process has to wait.

→ In multiprogramming system, the process should request the device to O.S

→ Device can be physical ( hard disk )
virtual ( virtual RAM ).

(4) Information Management.

System call include

get date, time, system data
set date, time system data
get file, device, Process attribute
set file, disk, Process attribute.

→ The user can fetch the system data, date time like this using such system call.
→ current date, time will be fetched
→ available free memory will be fetched

D D M M Y Y Y Y

(5) Communication management

↳ The communication blw user & OS is done smoothly using this management.

System call include: create a commu link, delet a link
snd & reciev information
Status of information
add / remov remote device.

↳ The communication link can be created blw two process for smooth performace using above system call.

↳ The inter process communication happen in two ways.
→ Shared memory
→ message passing.

→ Shared memory has a common location at which information are dumped & taken

→ message passing include a Kernal and each time system call is used to read & write data.

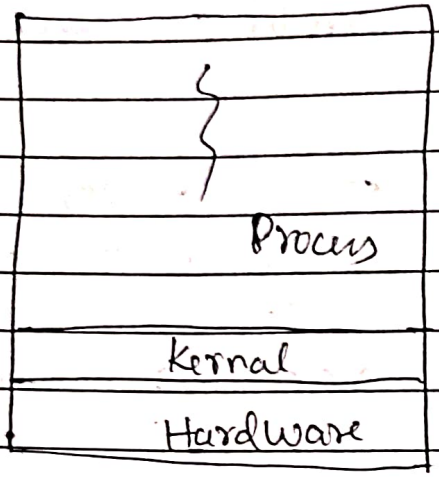
(6) Protection

↳ The OS carefully allow and block users access to process for accessing, file, device, information and other resources of system.

D D M M Y Y Y Y

2)

a) Virtual machine.

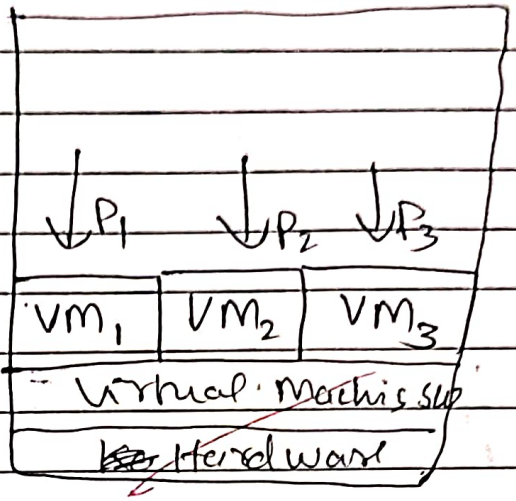



Non virtual Machine.

in general, this system contain RAM to hold and excute process's and

when one process is running another process has to wait till the first process complete or leave CPU as require I/O operation.

→ The CPU utilization is not utilized to the maximu in this way. and if any Process P5, with BT=1 ms come it shld wait for all before 4 Pros. to excute, Only then it get CPU Although it require only 1 ms to complete Job, WT was more and this makes this system in efficient.




Virtual Machine contains the software which exactly copies the below Hardware System & duplicate it and create many virtual machine (VM1, VM2,

Here, the CPU and memory, bus, I/O and all other peripherals are divided among the VM's and when $P_1$ is excuting in $VM_1$, $P_2$ can excute in $VM_2$ ... soon, like this Symententatially many process can be excuted at once.

CPU utilization is utilized to MAX in this manner, CPU never stay idle and even other services are also utilized properly like I/O, file access, memory allocation, and including the CPU utilization. Benefits ?

example for virtual Machine: Virtual box

Virtual box is software that allows user to run diffrent OS on current OS.

A user logined in with Windows OS can open virtual box & on a new machine say linux.

→ By this resources & services of system are utilized maximu

D D M M Y Y Y Y

Quiz:

1) a) fork

2) a) when procs

3) b) comm b/w two procs

4) b) Program count

5) b) 5